

 POLITÉCNICO DE LEIRIA	Organisation: IPLeiria	Date: 10/02/2024
	Customer: NA	Project: Master Thesis
	Team: NA	Ref: MEAU-DataAcqBoard

MCU Mapping Report

1 Job data

Type	Location	Reference
MCU mapping report	Leiria	MEAU-DataAcqBoard

Done by	Start date	End date
Joao Gaspar	10/02/2024	10/02/2024

Brief

The current document provides a summary of the decisions made to generate the MCU pin mapping, considering all requirements established in the Requirements Report.

2 Requirements

2.1 Develop microcontroller based module (familiarity, ease of deployment, enough processing power)

Solution Selected **ESP32** as processing unit for the module. Robust hardware and firmware design support.

2.2 Capable of acquiring at least 10 analog signals (NTCs) and 1 digital signal (flow meter)

Solution Resorted to **both ESP32 GPIOs and ADC expander** to acquire a total of 13 inputs.

- 10 NTC sensor inputs (analog)
- 1 Flow Meter input (digital)
- 1 Generic input (analog/digital)
- 1 User START button (digital)

To read said signals the following strategy was adopted:

1. Deployed MCP3008 as the ADC expander. This device will read 8 NTC. This device will require implementing SPI bus.
2. Remaining inputs (both analog and digital) to be mapped to ESP32. Map all said inputs to input only IOs.

2.3 Analog signal voltage reading error must not exceed 5%

Solution All signals to be read will be condition to range between 0 - 3.3V. This means that to meet this requirement all read signal voltages must be within $\pm 0.165V$ of the real value.

1. Based on previous experience using the ESP's ADC it is expected that a correction value will need to be applied.
2. The ADC expander has a resolution of 10bits, resulting in a resolution of 0.0032V per ADC bit. It is expected that an error less than 5% will be easily achievable with minimal calibration.

2.4 Capable of storing the acquired data in permanent storage at 2Hz

Solution Deploy SD Card for permanent log file storage. This requires adding a second CS pin to control data flow to the SD Card.

2.5 Capable of displaying acquired data to user in a timely manner (test safety control in real time)

Solution The flashing UART interface used by PlatformIO will be capable of meeting this requirement if new data lines are kept at least 100ms apart. This data frequency (10Hz) is enough to keep track of real time test conditions.

2.6 Hardware to enable manual control of data acquisition start/stop

Solution Deploy a user button mapped to one of the input-only ESP pins.

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3 Optional Features	
3.1	Implementation of CAN Bus to allow future use as datalogger of data generated elsewhere
Solution	Deploy CAN support hardware. 1. TJA1441ATK-0Z CAN transceiver. 2. RC filter between CAN H and CAN L lines.

3.2	Power outputs (+12V) to allow board to drive cooling accessories. Water Pump output is to be used to generate a PWM signal to set pump speed only. (Output current rating: 1A) Fan output is to be used to supply power to a fan with a current rating of 0.3A. (Output current rating: 1A) Extra power output may be used to power other loads. (Output current rating: 2A)
Solution	Deploy MOSFET+Driver setup for each power output to enable faster response when using PWM to drive these outputs. 1. MCP1407-E/MF MOSFET driver 2. PMV15UNEAR MOSFET (rated for 6A of current)

4 MCU Mapping

Taking into consideration the decisions made in points 2 and 3, the MCU's pins were mapped to specific functions. These are exposed in the tables below.

Table 1 – MCU mapping color code caption

Color code	Meaning
	Generic IO
	Input only
	Flash only pins

Table 2 – MCU mapping

ESP pad	Designator	Functionality/Peripheral
1	VDDA	Power
2	LNA_IN	Wi-Fi/Bluetooth Antenna Signal
3	VDDA3P3	Power
4	VDDA3P3	Power
5	SENSOR_VP (IO36)	Extra Filtered Signal (MCU_AIN1)
6	SENSOR_CAPP (IO37)	Unconnected
7	SENSOR_CAPN (IO38)	Unconnected
8	SENSOR_VN (IO39)	Unconnected
9	EN	ESP Reset Pin
10	IO34	NTC9 Filtered Signal (MCU_AIN2)
11	IO35	Flow Meter Filtered Signal (MCU_AIN3)
12	IO32	NTC8 Filtered Signal (MCU_AIN0)
13	IO33	User Button Digital Input (MCU_D3)
14	IO25	SPI (SD Card CS pin)
15	IO26	Water Pump PWM - Digital Output (MCU_D0)
16	IO27	Test Point
17	IO14	Unconnected
18	IO12	Test Point
19	VDD3P3_RTC	Power
20	IO13	CAN (CAN_RX - TJA1441ATK-0Z pin 4)
21	IO15	CAN (CAN_TX - TJA1441ATK-0Z pin 1)
22	IO2	Test Point
23	IO0	ESP Boot pin
24	IO4	General Purpose LED
25	IO16	Unconnected
26	VDD_SDIO	Power
27	IO17	Unconnected
28	SD2	Unconnected
29	SD3	Unconnected
30	CMD	Unconnected
31	CLK	Unconnected
32	SD0	Unconnected
33	SD1	Unconnected
34	IO5	SPI (ADC Expander CS pin)
35	IO18	SPI (SCLK pin)
36	IO23	SPI (MOSI pin)

4 MCU Mapping (continued)

Table 2 – MCU mapping (continued)

37	VDD3P3_CPU	Power
38	IO19	SPI (MISO pin)
39	IO22	FAN Power - Digital Output (MCU_D1)
40	U0RXD	UART1 (PlatformIO flashing interface)
41	U0TXD	UART1 (PlatformIO flashing interface)
42	IO21	Extra Power Output - Digital Output (MCU_D2)
43	VDDA	Power
44	XTAL_N_NC	Unconnected
45	XTAL_P_NC	Unconnected
46	VDDA	Power
47	CAP2_NC	Unconnected
48	CAP1_NC	Unconnected
49	GND	Ground pad

5 Summary

The pin mapping allocates:

- 5 inputs via ESP32 ADC
- 8 analog inputs via MCP3008 SPI ADC
- 3 digital outputs for power control
- 3 communication interfaces (SPI, UART, CAN)

All mandatory requirements are addressed with 1 extra output, 1 extra input and 3 ESP32 pins accessible via test points, available for future expansion.